BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

SLEEP, BIOLOGICAL RHYTHMS, AND ANXIETY IN THE PERINATAL PERIOD: A SYSTEMATIC REVIEW PROTOCOL

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046767
Article Type:	Protocol
Date Submitted by the Author:	09-Nov-2020
Complete List of Authors:	Cunningham, Jasmyn; McMaster University Michael G DeGroote School of Medicine Qureshi, Aljeena; McMaster University Health Sciences Honours Program Green, Sheryl; McMaster University, Department of Psychiatry and Behavioural Neurosciences; St. Joseph's Healthcare Hamilton, Women's Health Concerns Clinic Frey, Benicio; McMaster University, Department of Psychiatry and Behavioural Neurosciences; St. Joseph's Healthcare Hamilton, Women's Health Concerns Clinic; Mood Disorders Program Slyepchenko, Anastasiya; McMaster University, Department of Psychiatry and Behavioural Neurosciences,
Keywords:	Anxiety disorders < PSYCHIATRY, SLEEP MEDICINE, MENTAL HEALTH, Maternal medicine < OBSTETRICS, PRIMARY CARE, Adult psychiatry < PSYCHIATRY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

SLEEP, BIOLOGICAL RHYTHMS, AND ANXIETY IN THE PERINATAL PERIOD: A SYSTEMATIC REVIEW PROTOCOL

Jasmyn E. A. Cunningham[1*], Aljeena R. Qureshi[2], Sheryl Green[3,4], Benicio N. Frey[3,4,5], & Anastasiya Slyepchenko[3]

[1]Michael G. DeGroote School of Medicine, McMaster University, Hamilton, Canada
[2]Health Sciences Honours Program, McMaster University, Hamilton, Canada
[3]Department of Psychiatry and Behavioural Neurosciences, McMaster University, Hamilton, Canada
[4]Women's Health Concerns Clinic, St. Joseph's Healthcare Hamilton, Hamilton, Canada
[5] Mood Disorders Program, St. Joseph's Healthcare Hamilton, Hamilton, Canada

*Corresponding author: Cunningham.jasmyn06@gmail.com

PROSPERO Registration: 200166

Word Count: 2032

For peer review only

Abstract

Introduction: Pregnancy and new parenthood is an exciting time, but also a stressful life event that can predispose to perinatal mental health challenges. Perinatal anxiety is one such challenge, and is an important contributor to maternal distress and other negative outcomes. Sleep and biological rhythms are often disrupted in the perinatal period. These disruptions have been associated with postpartum depression, and in some cases with perinatal anxiety. However, the literature concerning the association with perinatal anxiety is inconsistent and may be limited by methodological shortcomings. To our knowledge, there has been no comprehensive review published characterizing the relationships between sleep, biological rhythms, and perinatal anxiety to date. In this systematic review, we will summarize the current state of the literature concerning these relationships, allowing us to highlight gaps in current literature, and potentially inform clinical understanding of perinatal anxiety, sleep, and biological rhythms.

Methods and Analysis: Primary research articles will be eligible for inclusion if they assess perinatal anxiety using validated criteria (self-report or diagnostic), assess sleep and biological rhythms in the perinatal period, include >4 participants, and meet other inclusion/exclusion criteria. We will conduct comprehensive searches of MEDLINE, PsycINFO, Embase, and CINAHL, with coverage spanning from database conception to search date. Key search concepts include 1) the perinatal period, 2) sleep/biological rhythms, and 3) anxiety. Risk of Bias will be evaluated using the Cochrane Risk of Bias Tool. Study data will be narratively synthesized, with quantitative synthesis included if possible. When relevant, strength of evidence will be assessed using GRADE criteria, and potential publication bias will be assessed.

Ethics and Dissemination: Research ethics approval is not required. Study results will be reported according to PRISMA guidelines. Results will be disseminated to relevant stakeholders as conference presentation(s) and submitted for publication in a peer-reviewed journal.

Registration Details: PROSPERO: 200166.

(299 words)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and Limitations of this Study

- Broad search and inclusion criteria will allow for a comprehensive review characterizing the relationships between sleep, biological rhythms, and perinatal anxiety, which will highlight gaps in current understanding of these relationships.
- Given the state of the literature, the authors predict few included studies per research question and a large degree of heterogeneity between them (e.g. in methods of assessment of anxiety, sleep, and biological rhythms), which will limit the ability for quantitative analysis.
- In this study we have defined 'perinatal' as pregnancy and up to 12 months post-partum, as opposed to several weeks postpartum, as many women will continue to experience disruptions and mental health challenges for this more extended period.

For peer review only

1. INTRODUCTION

1.1 Rationale

While pregnancy and new parenthood is often regarded as an exciting and positive time, it is also a stressful life event [1], with many accompanying changes and disruptions that can predispose new parents to perinatal mental health challenges. Depression is the most studied mental disorder occurring in this period, affecting as many as 19% of women[1] during the first three months postpartum [2], and potentially contributing to negative outcomes not only for mothers but also for their family [3]. However, despite receiving less research attention, perinatal anxiety is also an important contributor to maternal distress [4], and often co-occurs with depression [5, 6]. While not formally acknowledged as a disorder in the DSM-5 [7], postpartum anxiety has a prevalence of approximately 15-18% (self-reported symptoms) and 9-10% (diagnosed) in developed countries [6], and is associated with negative outcomes [8, 9]. Similarly, the prevalence of anxiety symptoms during pregnancy ranges from 18-24% (self-reported) and 15-18% (clinical diagnosis) [6].

Accumulating evidence suggests that perinatal anxiety may negatively impact outcomes in mothers and their infants. In mothers, perinatal anxiety may affect not only psychosocial outcomes, such as maternal self-confidence and body image, but also obstetric outcomes, such as sick leave during pregnancy, increased visits to obstetricians, anemia, delivery through C-sections, and delivery prior to term [10-12]. Importantly, anxiety disorders during the perinatal period also appear to negatively influence infant outcomes, including birth weight, Apgar scores, parenting behavior, the relationship between mothers and their infants (including bonding – a key process in the development of a child), and excessive crying in infants [10, 13-16]. Emerging evidence suggests that perinatal anxiety disorders are predictive of children's emotional and behavioural disturbances at 4 years of age [17]. Thus, prevention and treatment of perinatal anxiety can potentially have a positive impact on both the mother and her child.

Sleep and biological rhythms are also commonly disrupted in the perinatal period [18-21]. In the general population, disturbed sleep has been associated with diabetes, obesity, metabolic syndrome, chronic pain, hypertension, and heart disease [22-24], and sleep deprivation can contribute to changes in immune and endocrine functioning, as well as cognition and aspects of attention [25-28]. Specifically, during pregnancy, poor quality of sleep is associated with negative outcomes such as increased risk of suicidal ideation, preterm birth, low birth weight, fetal growth restriction, and gestational diabetes [29-31].

Sleep disruption has also been associated with postpartum depression [21, 32], and in some cases with perinatal anxiety [32, 33]. However, the literature concerning the association with perinatal anxiety is inconsistent. For example, in one study, women who reported poor sleep

[1] The authors use terms such as "woman" and "mother," as the majority of published literature has studied women in the perinatal period. However, the authors acknowledge that individuals other than women give birth, and that further gender-inclusive research on the topic is required.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

had higher self-reported anxiety symptoms at six-months postpartum [32]. Similarly, another study reported that anxiety was associated with subjective, but not objective, sleep changes in the third trimester [34]. In contrast, Tham and colleagues showed that postnatal depression, but not anxiety, is associated with poor sleep quality during pregnancy [35]. Additionally, fewer studies have focused on the relationships between perinatal anxiety and biological rhythms other than sleep, although perinatal alterations in biological rhythms have been linked to negative mood outcomes [20, 21, 36]. Published literature concerning sleep, biological rhythms, and perinatal anxiety in many cases may also be limited by small sample sizes, heterogeneity and other methodological shortcomings [21]. Despite the negative outcomes associated with both anxiety and disturbed sleep during the perinatal period [4, 8, 9, 29-31], to our knowledge, there has been no comprehensive review published characterizing the relationships between these concepts to date. In this systematic review, we will summarize the current state of the literature concerning the relationships between sleep, biological rhythms, and perinatal anxiety, which will allow us to highlight gaps in the current understanding of these phenomena, as well as potentially inform clinical understanding of these constructs.

1.2 Objectives

Our aim is to elucidate the interactions between sleep, biological rhythms, and anxiety in the perinatal period (including pregnancy and up to 12 months post-partum). To do so, this systematic review will address the following research questions:

1) How do sleep and biological rhythms in the perinatal period affect the onset and course of perinatal anxiety, and vice versa?
2) How do sleep and biological rhythms in the perinatal period affect the outcomes of perinatal anxiety?
3) How does perinatal anxiety affect outcomes of sleep and biological rhythm disruptions?
4) Are there any chronotherapeutic treatments for anxiety that have been administered in the perinatal period?
5) What are the effects of treatment for perinatal anxiety on sleep and circadian rhythms? Are there any treatments for perinatal anxiety targeting sleep that have been implemented?

2. METHODS AND ANALYSIS

This protocol and the reporting of results will be based on PRISMA guidelines [37]. There has been no patient or public involvement in the design of this study.

2.1 Eligibility Criteria

Studies will be eligible for inclusion if they meet the following criteria: a) assess perinatal anxiety using a validated self-report tool or diagnostic criteria (including anxiety symptoms or disorders; or Post-Traumatic Stress Disorder, Obsessive-Compulsive Disorder, or their symptoms); b) assess sleep or biological rhythms in the perinatal period, either subjectively or objectively (including clinical questionnaires, clinical interviews, actigraphy, polysomnography, etc.); c) published in or translated to English; and d) primary research article with > 4 human participants. Studies will be excluded if they: a) include jet lag as their sole sleep disturbance; b)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

include duplicate data from a database or other included study; c) do not distinguish adequately between anxiety and symptoms of other mental disorders (such as depression).

2.2 Information Sources and Search Strategy

Comprehensive searches of MEDLINE, PsycINFO, Embase and CINAHL will be conducted to identify relevant articles for inclusion. Coverage spans from 1946 (MEDLINE), 1806 (PsycINFO), 1974 (Embase), and 1981 (CINAHL) to search date. The search strategy will include predetermined keywords and subject headings appropriate to each database (MeSH for Medline, etc.). See Appendix 1 for terms and sample search strategy. Key concepts include 1) the perinatal period (pregnancy and/or up to 12 months postpartum), 2) sleep and/or biological rhythms, and 3) anxiety. No search limits will be applied.

2.3 Screening and Data Extraction

Citation lists from searches will be imported to EndNote version X9 for de-duplication. Following de-duplication, citations and data will be managed using Covidence. Titles and abstracts will be independently screened by two reviewers (JEAC, ARQ) for potential relevance. Following this, full text of potentially relevant articles will be obtained and independently assessed for inclusion (JEAC/ARQ). In cases of disagreement at either stage, a third reviewer (AS) will assess the article for potential inclusion. Two reviewers (JEAC/ARQ) will independently extract data using a standardized, electronic data collection form embedded within Covidence. Data will be reviewed for clarity and consistency by a third reviewer (AS), and reconciled into one document.

Data will be extracted for the following variables: 1) assessment/diagnosis of sleep and/or biological rhythms and associated disturbances, and related information such as prevalence, course, treatment outcomes, etc.; 2) assessment/diagnosis of anxiety, and related information such as prevalence, course, treatment outcomes, etc.; 3) characteristics of included participants, including pregnancy- and childbirth-related information; 4) study details, such as year of publication, number of included participants, comparator groups, methods of recruitment, timing and type of assessments, cut-off points, statistical analysis methods, and main findings. The first two domains of variables related to sleep, biological rhythms, and anxiety, are the main outcomes of interest. It should be noted that methods of measuring sleep and biological rhythms as well as methods of assessment of anxiety and related disorders will likely vary substantially between the studies. Current methods of assessment of sleep and biological rhythms are varied and may include subjective questionnaires (e.g. the Pittsburgh Sleep Quality Index [38], the Insomnia Severity Index [39], the Biological Rhythms Interview of Assessment in Neuropsychiatry [40]), actigraphy, polysomnography, measurement of body temperature, and fluctuation in hormones such as melatonin among others. Though there is no perinatal-specific anxiety diagnosis included in the DSM-5, a diagnosis of anxiety may be established using validated clinical interviews (e.g. the Mini International Neuropsychiatric Interview, the Structured Clinical Interview for the DSM-5), and clinical questionnaires may be used to assess symptom severity (e.g. the State-Trait Anxiety Inventory, the Generalized Anxiety Disorder-7, the Yale-Brown Obsessive Compulsive Scale, and the Depression, Anxiety and Stress Scale). Some perinatal-specific anxiety scales exist, including the Perinatal Obsessive-

For peer review only

Compulsive Scale, the anxiety subscale of the Edinburgh Postnatal Depression Scale, and the Perinatal Anxiety Screening Scale.

2.4 Data Synthesis and Evaluation

Risk of Bias will be evaluated using the Cochrane Risk of Bias (ROB) Tool [41]. Risk of Bias will be assessed at the level of the study. Study data will be narratively synthesized. Quantitative synthesis will be included if possible, however given the state of the literature on the topic, the authors predict a large degree of study heterogeneity and a low number of relevant studies per research question, which may limit the appropriateness of quantitative methods. When relevant, strength of evidence will be assessed using GRADE criteria, and potential publication bias will be assessed.

3. ETHICS AND DISSEMINATION

Research ethics approval is not required. Results will be disseminated to relevant stakeholders as conference presentation(s) and submitted for publication in a peer-reviewed journal.

4. DISCUSSION

Understanding the complex relationships of perinatal anxiety with sleep and biological rhythms will allow researchers and clinicians to begin to assess the importance of perinatal sleep and biological rhythms as risk factors and etiological components of anxiety during this period. As part of this review, we aim to provide a comprehensive and detailed overview of the existing literature on this topic, and to provide a foundation for future studies aiming to fill the gaps in current understanding of these relationships. Notably, a previous systematic review identified comorbid sleep disorders to be a risk factor for the first-time onset of anxiety during the perinatal period [42], indicating an important role of sleep in the development of anxiety during this time. If a substantial relationship is identified among sleep, biological rhythms and anxiety during the perinatal period, it is possible that sleep and biological rhythms may offer targets for pharmacological or psychotherapeutic treatment and prevention of anxiety during this critical period in the lives of women and their families.

4.1 Strengths and Limitations

Given the state of the current literature, the authors predict that few studies will address each of the research questions included in this study. Additionally, a large degree of heterogeneity is expected in the assessment of sleep and biological rhythms and of perinatal anxiety in the literature, due to the diverse range of methods currently used to assess these outcomes (e.g. actigraphy, polysomnography, subjective questionnaires, clinical interviews). This may limit our ability to conduct a quantitative analysis.

However, a strength of our protocol lies in including a broad search and inclusion criteria which will allow us to comprehensively describe these phenomena, and to highlight the current literature describing their interrelationships. Another strength of our protocol is the definition of “perinatal” as including pregnancy and up to 12 months post-partum, as opposed to merely pregnancy and several weeks immediately following childbirth. Many women continue to experience disruptions in their mental health related to perinatal-specific stressors in this

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

extended period, and thus a more comprehensive characterization has the potential to benefit women throughout a longer duration following childbirth.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

REFERENCES

1. Holmes, T.H. and R.H. Rahe, *The Social Readjustment Rating Scale.* J Psychosom Res, 1967. **11**(2): p. 213-8.
2. Gavin, N.I., et al., *Perinatal depression: a systematic review of prevalence and incidence.* Obstet Gynecol, 2005. **106**(5 Pt 1): p. 1071-83.
3. Weissman, M.M., *Postpartum Depression and Its Long-term Impact on Children: Many New Questions.* JAMA Psychiatry, 2018. **75**(3): p. 227-228.
4. Miller, R.L., J.F. Pallant, and L.M. Negri, *Anxiety and stress in the postpartum: is there more to postnatal distress than depression?* BMC Psychiatry, 2006. **6**: p. 12.
5. Falah-Hassani, K., R. Shiri, and C.L. Dennis, *Prevalence and risk factors for comorbid postpartum depressive symptomatology and anxiety.* J Affect Disord, 2016. **198**: p. 142-7.
6. Dennis, C.L., K. Falah-Hassani, and R. Shiri, *Prevalence of antenatal and postnatal anxiety: systematic review and meta-analysis.* Br J Psychiatry, 2017. **210**(5): p. 315-323.
7. American Psychiatric Association, T., *Diagnostic and Statistical Manual of Mental Disorders (5th ed.)*. 2013, Arlington, VA: American Psychiatric Publishing.
8. Glover, V., *Maternal depression, anxiety and stress during pregnancy and child outcome; what needs to be done.* Best Pract Res Clin Obstet Gynaecol, 2014. **28**(1): p. 25-35.
9. Stein, A., et al., *Effects of perinatal mental disorders on the fetus and child.* The Lancet, 2014. **384**(9956): p. 1800-1819.
10. Goodman, J.H., G.R. Watson, and B. Stubbs, *Anxiety disorders in postpartum women: A systematic review and meta-analysis.* J Affect Disord, 2016. **203**: p. 292-331.
11. Andersson, L., et al., *Implications of antenatal depression and anxiety for obstetric outcome.* Obstet Gynecol, 2004. **104**(3): p. 467-76.
12. Banhidy, F., et al., *Association between maternal panic disorders and pregnancy complications and delivery outcomes.* Eur J Obstet Gynecol Reprod Biol, 2006. **124**(1): p. 47-52.
13. Petzoldt, J., et al., *Maternal anxiety disorders predict excessive infant crying: a prospective longitudinal study.* Arch Dis Child, 2014. **99**(9): p. 800-6.
14. Murray, L., et al., *The effects of maternal social phobia on mother-infant interactions and infant social responsiveness.* J Child Psychol Psychiatry, 2007. **48**(1): p. 45-52.
15. Berle, J.O., et al., *Neonatal outcomes in offspring of women with anxiety and depression during pregnancy. A linkage study from The Nord-Trondelag Health Study (HUNT) and Medical Birth Registry of Norway.* Arch Womens Ment Health, 2005. **8**(3): p. 181-9.
16. Grigoriadis, S., et al., *Maternal Anxiety During Pregnancy and the Association With Adverse Perinatal Outcomes: Systematic Review and Meta-Analysis.* J Clin Psychiatry, 2018. **79**(5).
17. O'Connor, T.G., et al., *Maternal antenatal anxiety and children's behavioural/emotional problems at 4 years. Report from the Avon Longitudinal Study of Parents and Children.* Br J Psychiatry, 2002. **180**: p. 502-8.
18. Facco, F.L., et al., *Sleep disturbances in pregnancy.* Obstet Gynecol, 2010. **115**(1): p. 77-83.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

19. Signal, T.L., et al., *Sleep duration and quality in healthy nulliparous and multiparous women across pregnancy and post-partum.* Aust N Z J Obstet Gynaecol, 2007. **47**(1): p. 16-22.
20. Sharkey, K.M., T.B. Pearlstein, and M.A. Carskadon, *Circadian phase shifts and mood across the perinatal period in women with a history of major depressive disorder: a preliminary communication.* J Affect Disord, 2013. **150**(3): p. 1103-8.
21. Gallaher, K.G.H., et al., *The Role of Circadian Rhythms in Postpartum Sleep and Mood.* Sleep Med Clin, 2018. **13**(3): p. 359-374.
22. Foley, D., et al., *Sleep disturbances and chronic disease in older adults: results of the 2003 National Sleep Foundation Sleep in America Survey.* J Psychosom Res, 2004. **56**(5): p. 497-502.
23. Hall, M.H., et al., *Self-reported sleep duration is associated with the metabolic syndrome in midlife adults.* Sleep, 2008. **31**(5): p. 635-43.
24. Gottlieb, D.J., et al., *Association of usual sleep duration with hypertension: the Sleep Heart Health Study.* Sleep, 2006. **29**(8): p. 1009-14.
25. Cunningham, J.E.A., et al., *Acute Sleep Restriction Has Differential Effects on Components of Attention.* Front Psychiatry, 2018. **9**: p. 499.
26. Van Dongen, H.P., et al., *The cumulative cost of additional wakefulness: dose-response effects on neurobehavioral functions and sleep physiology from chronic sleep restriction and total sleep deprivation.* Sleep, 2003. **26**(2): p. 117-26.
27. Spiegel, K., R. Leproult, and E. Van Cauter, *Impact of sleep debt on metabolic and endocrine function.* The Lancet, 1999. **354**(9188): p. 1435-1439.
28. Spiegel, K., J.F. Sheridan, and E. Van Cauter, *Effect of sleep deprivation on response to immunization.* JAMA, 2002. **288**(12): p. 1471-2.
29. Cai, S., et al., *Sleep Quality and Nocturnal Sleep Duration in Pregnancy and Risk of Gestational Diabetes Mellitus.* Sleep, 2017. **40**(2).
30. Gelaye, B., et al., *Association of poor subjective sleep quality with suicidal ideation among pregnant Peruvian women.* Gen Hosp Psychiatry, 2015. **37**(5): p. 441-7.
31. Micheli, K., et al., *Sleep patterns in late pregnancy and risk of preterm birth and fetal growth restriction.* Epidemiology, 2011. **22**(5): p. 738-44.
32. Okun, M.L., et al., *Poor sleep quality increases symptoms of depression and anxiety in postpartum women.* J Behav Med, 2018. **41**(5): p. 703-710.
33. Lawson, A., et al., *The relationship between sleep and postpartum mental disorders: A systematic review.* J Affect Disord, 2015. **176**: p. 65-77.
34. Bei, B., et al., *Subjective perception of sleep, but not its objective quality, is associated with immediate postpartum mood disturbances in healthy women.* Sleep, 2010. **33**(4): p. 531-8.
35. Tham, E.K., et al., *Associations between poor subjective prenatal sleep quality and postnatal depression and anxiety symptoms.* J Affect Disord, 2016. **202**: p. 91-4.
36. Krawczak, E.M., et al., *Do changes in subjective sleep and biological rhythms predict worsening in postpartum depressive symptoms? A prospective study across the perinatal period.* Arch Womens Ment Health, 2016. **19**(4): p. 591-8.
37. Moher, D., et al., *Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement.* PLoS Med, 2009. **6**(7): p. e1000097.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

38. Buysse, D.J., et al., *The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research.* Psychiatry Res, 1989. **28**(2): p. 193-213.
39. Bastien, C.H., A. Vallieres, and C.M. Morin, *Validation of the Insomnia Severity Index as an outcome measure for insomnia research.* Sleep Medicine, 2001. **2**: p. 297-307.
40. Giglio, L.M., et al., *Development and use of a biological rhythm interview.* J Affect Disord, 2009. **118**(1-3): p. 161-5.
41. Higgins, J.P., et al., *The Cochrane Collaboration's tool for assessing risk of bias in randomised trials.* BMJ, 2011. **343**: p. d5928.
42. Furtado, M., et al., *Risk factors of new onset anxiety and anxiety exacerbation in the perinatal period: A systematic review and meta-analysis.* J Affect Disord, 2018. **238**: p. 626-635.

For peer review only

AUTHOR CONTRIBUTIONS

JEAC: Contributed to study design; drafted and critically revised research protocol
ARQ: Contributed to study design; drafted and critically revised research protocol
SG: Critical revision of research protocol
BNF: Contributed to study idea & critical revision of research protocol; provided research supervision
AS*: Conceived of study idea; contributed to study design & critical revision of research protocol; provided research supervision
*Review quality guarantor

FUNDING STATEMENT

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

COMPETING INTERESTS STATEMENT

AS and BNF have received research funding from Pfizer. JEAC, ARQ, and SG have no competing interests to declare.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

APPENDIX 1

Table 1: Key concepts, terms, and example search strategy for MEDLINE database search.

Key Concept	Perinatal Period	Anxiety	Sleep and Biological Rhythms
Free-text terms	• Perinatal • Postpartum • Post-partum • Post-partum period • Postnatal • Perinatal period • Peripartum • Pregnan* • Antenatal • Gestation • Parity • Gravid* • Puerperium	• PTSD • Post-traumatic Stress disorder • Post Traumatic Stress Disorder • GAD • Generalized Anxiety Disorder • General Anxiety • OCD • Obsessive compulsive disorder • Obsessive-compulsive disorder • Perinatal Anxiety • Anxiety • Social anxiety • Panic Disorder • Phobia	• Insomnia • Circadian • Sleep phase advance • Sleep phase delay • Sleep loss • Sleep difficult* • Sleep reactiv* • Maternal sleep • Nyctohemeral • Nychtohemeral • Diurnal • twenty-four hour rhythm • Twenty four hour rhythm • 24 hour rhythm • day night • Actiwatch • biological rhythm*
MeSH Subject Headings	• Pregnancy • Pregnancy, High-Risk • Pregnancy, Multiple • Pregnancy, Unplanned • Pregnancy, Unwanted • Gravidity • Parity • Parturition • Peripartum Period • Postpartum Period • Mothers • Single Parent • Pregnant Women • Child Rearing	• Obsessive Behavior • Stress, Psychological • Hoarding • Impulsive Behavior • Compulsive Behavior • Anxiety • Catastrophization • [Explode] Anxiety Disorders • "Trauma and Stressor Related Disorders" • Adjustment Disorders • Stress Disorders, Traumatic • Stress Disorders, Post-Traumatic	• Chronobiology Disorders • Sleep Disorders, Circadian Rhythm • Fatigue Syndrome, Chronic • Restless Legs Syndrome • [Explode] Sleep Wake Disorders • Sleep Hygiene • [Explode] Chronobiology Phenomena • [Explode] Sleep

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	• Maternal-Fetal Relations • Puerperal Disorders	• Stress Disorders, Traumatic, Acute • Hoarding Disorder	• [Explode] Accelerometry • [Explode] Chronotherapy • Phototherapy • Sleep Phase Chronotherapy
Example Search Strategy	• Terms within key concepts combined with OR • Key concept groups of terms combined with AND • Example *(truncated for clarity)*: (Perinatal OR Postpartum OR …) AND (PTSD OR Anxiety OR …) AND (Insomnia OR Circadian OR …)		

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	1
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments			

For peer review only

	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	12
Sponsor	#5b	Provide name for the review funder and / or sponsor	12
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	12
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	13
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	6

For peer review only

		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	7
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	7
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	7
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	7
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	7
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	7
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	7

The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist was completed on 09. November 2020 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

For peer review only

BMJ Open

SLEEP, BIOLOGICAL RHYTHMS, AND ANXIETY IN THE PERINATAL PERIOD: A SYSTEMATIC REVIEW PROTOCOL

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046767.R1
Article Type:	Protocol
Date Submitted by the Author:	14-Apr-2021
Complete List of Authors:	Cunningham, Jasmyn; McMaster University Michael G DeGroote School of Medicine Qureshi, Aljeena; McMaster University Health Sciences Honours Program Green, Sheryl; McMaster University, Department of Psychiatry and Behavioural Neurosciences; St. Joseph's Healthcare Hamilton, Women's Health Concerns Clinic Frey, Benicio; McMaster University, Department of Psychiatry and Behavioural Neurosciences; St. Joseph's Healthcare Hamilton, Women's Health Concerns Clinic; Mood Disorders Program Slyepchenko, Anastasiya; McMaster University, Department of Psychiatry and Behavioural Neurosciences,
Primary Subject Heading:	Mental health
Secondary Subject Heading:	Reproductive medicine, General practice / Family practice, Obstetrics and gynaecology
Keywords:	Anxiety disorders < PSYCHIATRY, SLEEP MEDICINE, MENTAL HEALTH, Maternal medicine < OBSTETRICS, PRIMARY CARE, Adult psychiatry < PSYCHIATRY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

SLEEP, BIOLOGICAL RHYTHMS, AND ANXIETY IN THE PERINATAL PERIOD: A SYSTEMATIC REVIEW PROTOCOL

Jasmyn E. A. Cunningham[1*], Aljeena R. Qureshi[2], Sheryl M. Green[3,4], Benicio N. Frey[3,4,5], & Anastasiya Slyepchenko[3]

[1]Michael G. DeGroote School of Medicine, McMaster University, Hamilton, Canada
[2]Health Sciences Honours Program, McMaster University, Hamilton, Canada
[3]Department of Psychiatry and Behavioural Neurosciences, McMaster University, Hamilton, Canada
[4]Women's Health Concerns Clinic, St. Joseph's Healthcare Hamilton, Hamilton, Canada
[5] Mood Disorders Program, St. Joseph's Healthcare Hamilton, Hamilton, Canada

[*]Corresponding author: Cunningham.jasmyn06@gmail.com

PROSPERO Registration: 200166

Word Count: 2331

For peer review only

Abstract

Introduction: Pregnancy and new parenthood is an exciting time, but also a stressful life event that can predispose to mental health challenges. Perinatal anxiety is one such challenge, and is an important contributor to parental distress and other negative outcomes. Sleep and biological rhythms are often disrupted in the perinatal period. These disruptions have been associated with postpartum depression, and in some cases with perinatal anxiety. However, the literature concerning the association with perinatal anxiety is inconsistent and may be methodologically limited. To our knowledge, there has been no comprehensive review published characterizing the relationships between sleep, biological rhythms, and perinatal anxiety and related disorders to date. In this systematic review, we will summarize the current state of the literature concerning these relationships, allowing us to highlight gaps and potentially inform clinical understanding of perinatal anxiety, sleep, and biological rhythms.

Methods and Analysis: Primary research articles will be eligible for inclusion if they assess perinatal anxiety or related disorders using validated criteria (self-report or diagnostic), assess sleep and biological rhythms in the perinatal period, include >4 participants, and meet other inclusion/exclusion criteria. We will conduct comprehensive searches of MEDLINE, PsycINFO, Embase, and CINAHL, with coverage spanning from database conception to search date (April 2021). Key search concepts include 1) the perinatal period, 2) sleep/biological rhythms, and 3) anxiety. Risk of Bias will be evaluated using the Cochrane Risk of Bias Tool. Data will be narratively synthesized, with quantitative synthesis included if possible. When relevant, strength of evidence will be assessed using GRADE criteria, and potential publication bias will be assessed.

Ethics and Dissemination: Research ethics approval is not required. Study results will be reported according to PRISMA guidelines. Results will be disseminated to relevant stakeholders as conference presentation(s) and submitted for publication in a peer-reviewed journal.

Registration Details: PROSPERO: 200166.

(300/300 words)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and Limitations of this Study

- Broad search and inclusion criteria will allow for a comprehensive review characterizing the relationships between sleep, biological rhythms, and perinatal anxiety and related disorders, which will highlight gaps in current understanding of these relationships.
- Given the state of the literature, the authors predict few included studies per research question and a large degree of heterogeneity between them (e.g. in methods of assessment of anxiety, sleep, and biological rhythms), which will limit the ability for quantitative analysis.
- In this study we have defined 'perinatal' as pregnancy and up to 12 months post-partum, as opposed to several weeks postpartum, as many of these individuals will continue to experience disruptions and mental health challenges for this more extended period.

For peer review only

1. INTRODUCTION

1.1 Rationale

While pregnancy and new parenthood is often regarded as an exciting and positive time, it is also a stressful life event [1], with many accompanying changes and disruptions that can predispose new parents and new parents to be to mental health challenges during pregnancy and the postpartum (the perinatal period). Depression is the most studied mental disorder occurring in this period, affecting as many as 19% of persons during the first three months postpartum [2], and potentially contributing to negative outcomes not only for the postpartum individual but also for their family [3]. However, despite receiving less research attention, perinatal anxiety is also an important contributor to parental distress [4], and often co-occurs with depression, making both disorders more difficult to treat [4-7]. Given its common co-occurrence with depression, anxiety is often studied in the context of this disorder; however, it is important to distinguish the differential impact of anxiety from depression given its respective contribution to distress as well as to better inform appropriate treatment options [4]. While not formally acknowledged as a disorder in the DSM-5 [8], postpartum anxiety has a prevalence of approximately 15-18% (self-reported symptoms) and 9-10% (diagnosed) in developed countries [6], and is associated with negative outcomes [9, 10]. Similarly, the prevalence of anxiety symptoms during pregnancy ranges from 18-24% (self-reported) and 15-18% (clinical diagnosis) [6].

Accumulating evidence suggests that perinatal anxiety may negatively impact outcomes in peripartum individuals and their infants. In peripartum individuals, perinatal anxiety may affect not only psychosocial outcomes, such as parental self-confidence and body image, but also obstetric outcomes, such as sick leave during pregnancy, increased visits to obstetricians, anemia, delivery through C-sections, and delivery prior to term [11-13]. Importantly, anxiety disorders during the perinatal period also appear to negatively influence infant outcomes, including birth weight, Apgar scores, parenting behavior, the relationship between mothers and their infants (including bonding – a key process in the development of a child), and excessive crying in infants [11, 14-17]. Emerging evidence suggests that perinatal anxiety disorders are predictive of children's emotional and behavioural disturbances at 4 years of age [18]. Thus, prevention and treatment of perinatal anxiety can potentially have a positive impact on both the parent and their child.

Although no longer formally categorized as anxiety disorders in the DSM-5, anxiety-related disorders such as such as obsessive-compulsive disorder (OCD) and post-traumatic stress disorder (PTSD) are also prevalent in the perinatal period and have been associated with poor health outcomes for mothers and infants. The existing literature shows that the perinatal period represents a time of increased risk for the development and exacerbation of OCD symptoms, with an estimated prevalence of 0.2-3.5% in pregnancy and up to 2.3-9% in the postpartum [19-21]. OCD during pregnancy is frequently associated with contamination fears regarding the fetus, and can present with compulsions such as excessive washing or cleaning rituals, diet restriction, and restriction of contact with others postnatally. Postnatal onset of

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

OCD has been associated with aggressive intrusion cognitions involving deliberately harming the baby and is associated with the avoidance of caregiving tasks such as bathing, or compulsive checking, for example that the infant hasn't been kidnapped by a stranger [19, 20]. While literature regarding PTSD during the perinatal period is relatively limited both in quantity and in methodological quality, PTSD during the postpartum period has also been associated with negative outcomes including low birth weight and decreased rates of breastfeeding [22].

Sleep and biological rhythms are also commonly disrupted in the perinatal period [23-26]. In the general population, disturbed sleep has been associated with diabetes, obesity, metabolic syndrome, chronic pain, hypertension, and heart disease [27-29], and sleep deprivation can contribute to changes in immune and endocrine functioning, as well as cognition and aspects of attention [30-33]. Poor sleep and the presence of sleep disorders are also associated with a lower quality of life, and interference with performance of important activities such as driving [34, 35]. Specifically, during pregnancy, poor quality of sleep is associated with negative outcomes such as increased risk of suicidal ideation, preterm birth, low birth weight, fetal growth restriction, and gestational diabetes [36-38].

Sleep disruption has also been associated with postpartum depression [26, 39], and in some cases with perinatal anxiety [39, 40]. However, the literature concerning the association with perinatal anxiety and related disorders is inconsistent. For example, in one study, women who reported poor sleep had higher self-reported anxiety symptoms at six-months postpartum [39]. Similarly, other studies have reported that anxiety was associated with subjective, but not objective, sleep changes in the third trimester [41], and higher levels of anxiety and symptoms of OCD were reported among individuals experiencing mid-pregnancy insomnia [21]. In contrast, Tham and colleagues showed that postnatal depression, but not anxiety, is associated with poor sleep quality during pregnancy [42]. Sleep difficulties are also common in PTSD [43], and sleep difficulties in the perinatal period have been associated with a worsening trajectory of posttraumatic stress across this same period [44]. Additionally, fewer studies have focused on the relationships between perinatal anxiety and biological rhythms other than sleep, although perinatal alterations in biological rhythms have been linked to negative mood outcomes [25, 26, 45]. Published literature concerning sleep, biological rhythms, and perinatal anxiety and related disorders in many cases may also be limited by small sample sizes, heterogeneity and other methodological shortcomings [26]. Despite the negative outcomes associated with both anxiety and disturbed sleep during the perinatal period [4, 9, 10, 36-38], to our knowledge, there has been no comprehensive review published characterizing the relationships between these concepts to date. In this systematic review, we will summarize the current state of the literature concerning the relationships between sleep, biological rhythms, and perinatal anxiety and related disorders, which will allow us to highlight gaps in the current understanding of these phenomena, as well as potentially inform clinical understanding of these constructs.

1.2 Objectives

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Our aim is to elucidate the interactions between sleep, biological rhythms, and anxiety and related disorders in the perinatal period (including pregnancy and up to 12 months post-partum). To do so, this systematic review will address the following research questions:

1) How do sleep and biological rhythms in the perinatal period affect the onset and course of perinatal anxiety and related disorders, and vice versa?
2) How do sleep and biological rhythms in the perinatal period affect the outcomes of perinatal anxiety and related disorders?
3) How do perinatal anxiety and related disorders affect outcomes of sleep and biological rhythm disruptions?
4) Are there any chronotherapeutic treatments that have been administered in the perinatal period to treat anxiety and related disorders, or sleep disturbances in individuals with anxiety and related disorders?
5) What are the effects of treatment for perinatal anxiety and related disorders on sleep and circadian rhythms? Are there any treatments for perinatal anxiety and related disorders targeting sleep that have been implemented? What were the effects of the implemented treatments?

2. METHODS AND ANALYSIS

This protocol and the reporting of results will be based on PRISMA guidelines [46].

2.1 Eligibility Criteria

Studies will be eligible for inclusion if they meet the following criteria: a) assess perinatal anxiety using a validated self-report tool or diagnostic criteria (including anxiety symptoms or disorders; or Post-Traumatic Stress Disorder, Obsessive-Compulsive Disorder, or their symptoms); b) assess sleep or biological rhythms in the perinatal period, either subjectively or objectively (including clinical questionnaires, clinical interviews, actigraphy, polysomnography, etc.); c) published in or translated to English; and d) primary research article with > 4 human participants. Studies will be excluded if they: a) include jet lag as their sole sleep disturbance; b) include duplicate data from a database or other included study; c) do not distinguish adequately between perinatal anxiety and symptoms of other mental disorders (such as depression).

2.2 Information Sources and Search Strategy

Comprehensive searches of MEDLINE, PsycINFO, Embase and CINAHL will be conducted to identify relevant articles for inclusion. Coverage spans from 1946 (MEDLINE), 1806 (PsycINFO), 1974 (Embase), and 1981 (CINAHL) to search date (April 2021). The search strategy will include predetermined keywords and subject headings appropriate to each database (MeSH for Medline, etc.). See Appendix 1 for terms and sample search strategy. Key concepts include 1) the perinatal period (pregnancy and/or up to 12 months postpartum), 2) sleep and/or biological rhythms, and 3) anxiety. No search limits will be applied.

2.3 Screening and Data Extraction

Citation lists from searches will be imported to EndNote version X9 for de-duplication. Following de-duplication, citations and data will be managed using Covidence. Titles and

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

abstracts will be independently screened by two reviewers for potential relevance. Following this, full text of potentially relevant articles will be obtained and independently assessed for inclusion. In cases of disagreement at either stage, a third reviewer will assess the article for potential inclusion. Two reviewers will independently extract data using a standardized, electronic data collection form embedded within Covidence. Data will be reviewed for clarity and consistency by a third reviewer, and reconciled into one document.

Data will be extracted for the following variables: 1) assessment/diagnosis of sleep and/or biological rhythms and associated disturbances, and related information such as prevalence, course, treatment outcomes, etc.; 2) assessment/diagnosis of anxiety or related disorder, and related information such as prevalence, course, treatment outcomes, etc.; 3) characteristics of included participants, including pregnancy- and childbirth-related information, as well as the presence of spousal/partner/familial support persons; 4) study details, such as year of publication, number of included participants, comparator groups, methods of recruitment, timing and type of assessments, cut-off points, statistical analysis methods, and main findings. The first two domains of variables related to sleep, biological rhythms, and anxiety, are the main outcomes of interest. It should be noted that methods of measuring sleep and biological rhythms as well as methods of assessment of anxiety and related disorders will likely vary substantially between the studies. Current methods of assessment of sleep and biological rhythms are varied and may include subjective questionnaires (e.g. the Pittsburgh Sleep Quality Index [47], the Insomnia Severity Index [48], the Biological Rhythms Interview of Assessment in Neuropsychiatry [49]), actigraphy, polysomnography, measurement of body temperature, and fluctuation in hormones such as melatonin among others. Though there is no perinatal-specific anxiety diagnosis included in the DSM-5, a diagnosis of an anxiety disorder may be established using validated clinical interviews (e.g. the Mini International Neuropsychiatric Interview, the Structured Clinical Interview for the DSM-5), and clinical questionnaires may be used to assess symptom severity (e.g. the State-Trait Anxiety Inventory, the Generalized Anxiety Disorder-7, the Yale-Brown Obsessive Compulsive Scale, and the Depression, Anxiety and Stress Scale). Some perinatal-specific anxiety scales exist, including the Perinatal Obsessive-Compulsive Scale, the anxiety subscale of the Edinburgh Postnatal Depression Scale, and the Perinatal Anxiety Screening Scale.

2.4 Data Synthesis and Evaluation

Risk of Bias will be evaluated using the Cochrane Risk of Bias (ROB) Tool [50]. Risk of Bias will be assessed at the level of the study. Study data will be narratively synthesized. Quantitative synthesis will be included if possible, however given the state of the literature on the topic, the authors predict a large degree of study heterogeneity and a low number of relevant studies per research question, which may limit the appropriateness of quantitative methods. Meta-regression, subgroup analysis, and/or random effects meta-analysis may be employed if appropriate given the high degree of expected heterogeneity. When relevant, strength of evidence will be assessed using GRADE criteria, and potential publication bias will be assessed.

2.5 Patient and Public Involvement

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

This study protocol was presented to the Community Advisory Committee consisting of former patients of the Women's Health Concerns Clinic at St. Joseph's Healthcare Hamilton (March 2021, Hamilton, Ontario, Canada). Two individuals with lived experience have read and provided feedback on this systematic review protocol.

3. ETHICS AND DISSEMINATION

Research ethics approval is not required. Results will be disseminated to relevant stakeholders as conference presentation(s) and submitted for publication in a peer-reviewed journal.

4. DISCUSSION

Understanding the complex relationships of perinatal anxiety with sleep and biological rhythms will allow researchers and clinicians to begin to assess the importance of perinatal sleep and biological rhythms as risk factors and etiological components of anxiety during this period. As part of this review, we aim to provide a comprehensive and detailed overview of the existing literature on this topic, and to provide a foundation for future studies aiming to fill the gaps in current understanding of these relationships. Notably, a previous systematic review identified comorbid sleep disorders to be a risk factor for the first-time onset of anxiety during the perinatal period [51], indicating an important role of sleep in the development of anxiety during this time. If a substantial relationship is identified among sleep, biological rhythms and anxiety during the perinatal period, it is possible that sleep and biological rhythms may offer targets for pharmacological or psychotherapeutic treatment and prevention of anxiety during this critical period in the lives of parents, parents to be, and their families.

4.1 Strengths and Limitations

Given the state of the current literature, the authors predict that few studies will address each of the research questions included in this study. Additionally, a large degree of heterogeneity is expected in the assessment of sleep and biological rhythms and of perinatal anxiety in the literature, due to the diverse range of methods currently used to assess these outcomes (e.g. actigraphy, polysomnography, subjective questionnaires, clinical interviews). This may limit our ability to conduct a quantitative analysis.

However, a strength of our protocol lies in including a broad search and inclusion criteria which will allow us to comprehensively describe these phenomena, and to highlight the current literature describing their interrelationships. Another strength of our protocol is the definition of "perinatal" as including pregnancy and up to 12 months post-partum, as opposed to merely pregnancy and several weeks immediately following childbirth. Many individuals continue to experience disruptions in their mental health related to perinatal-specific stressors in this extended period, and thus a more comprehensive characterization has the potential to benefit these individuals throughout a longer duration following childbirth.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

AUTHOR CONTRIBUTIONS

JEAC: Contributed to study design; drafted and critically revised research protocol
ARQ: Contributed to study design; drafted and critically revised research protocol
SMG: Critical revision of research protocol
BNF: Contributed to study idea & critical revision of research protocol; provided research supervision
AS*: Conceived of study idea; contributed to study design & critical revision of research protocol; provided research supervision
*Review quality guarantor

ACKNOWLEDGEMENTS

We would like to extend our gratitude to Christine Vanderveen and Brydie Huffman for providing feedback on the study protocol.

FUNDING STATEMENT

This work is sponsored in part by the Michael G. DeGroote School of Medicine McMaster Medical Student Research Excellence Award (MAC RES) and by an unrestricted fundraising gift from Shoppers Drug Mart (*Shoppers Run For Women*).

COMPETING INTERESTS STATEMENT

AS has received research funding from Pfizer. JEAC, ARQ, SMG and BNF have no competing interests to declare.

For peer review only

REFERENCES

1. Holmes, T.H. and R.H. Rahe, *The Social Readjustment Rating Scale.* J Psychosom Res, 1967. **11**(2): p. 213-8.
2. Gavin, N.I., et al., *Perinatal depression: a systematic review of prevalence and incidence.* Obstet Gynecol, 2005. **106**(5 Pt 1): p. 1071-83.
3. Weissman, M.M., *Postpartum Depression and Its Long-term Impact on Children: Many New Questions.* JAMA Psychiatry, 2018. **75**(3): p. 227-228.
4. Miller, R.L., J.F. Pallant, and L.M. Negri, *Anxiety and stress in the postpartum: is there more to postnatal distress than depression?* BMC Psychiatry, 2006. **6**: p. 12.
5. Falah-Hassani, K., R. Shiri, and C.L. Dennis, *Prevalence and risk factors for comorbid postpartum depressive symptomatology and anxiety.* J Affect Disord, 2016. **198**: p. 142-7.
6. Dennis, C.L., K. Falah-Hassani, and R. Shiri, *Prevalence of antenatal and postnatal anxiety: systematic review and meta-analysis.* Br J Psychiatry, 2017. **210**(5): p. 315-323.
7. Hawke, L.D., et al., *Engaging youth in research planning, design and execution: Practical recommendations for researchers.* Health Expect, 2018. **21**(6): p. 944–949.
8. American Psychiatric Association, T., *Diagnostic and Statistical Manual of Mental Disorders (5th ed.)*. 2013, Arlington, VA: American Psychiatric Publishing.
9. Glover, V., *Maternal depression, anxiety and stress during pregnancy and child outcome; what needs to be done.* Best Pract Res Clin Obstet Gynaecol, 2014. **28**(1): p. 25-35.
10. Stein, A., et al., *Effects of perinatal mental disorders on the fetus and child.* The Lancet, 2014. **384**(9956): p. 1800-1819.
11. Goodman, J.H., G.R. Watson, and B. Stubbs, *Anxiety disorders in postpartum women: A systematic review and meta-analysis.* J Affect Disord, 2016. **203**: p. 292-331.
12. Andersson, L., et al., *Implications of antenatal depression and anxiety for obstetric outcome.* Obstet Gynecol, 2004. **104**(3): p. 467-76.
13. Banhidy, F., et al., *Association between maternal panic disorders and pregnancy complications and delivery outcomes.* Eur J Obstet Gynecol Reprod Biol, 2006. **124**(1): p. 47-52.
14. Petzoldt, J., et al., *Maternal anxiety disorders predict excessive infant crying: a prospective longitudinal study.* Arch Dis Child, 2014. **99**(9): p. 800-6.
15. Murray, L., et al., *The effects of maternal social phobia on mother-infant interactions and infant social responsiveness.* J Child Psychol Psychiatry, 2007. **48**(1): p. 45-52.
16. Berle, J.O., et al., *Neonatal outcomes in offspring of women with anxiety and depression during pregnancy. A linkage study from The Nord-Trondelag Health Study (HUNT) and Medical Birth Registry of Norway.* Arch Womens Ment Health, 2005. **8**(3): p. 181-9.
17. Grigoriadis, S., et al., *Maternal Anxiety During Pregnancy and the Association With Adverse Perinatal Outcomes: Systematic Review and Meta-Analysis.* J Clin Psychiatry, 2018. **79**(5).
18. O'Connor, T.G., et al., *Maternal antenatal anxiety and children's behavioural/emotional problems at 4 years. Report from the Avon Longitudinal Study of Parents and Children.* Br J Psychiatry, 2002. **180**: p. 502-8.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

19. Frías, Á., et al., *Obsessive-compulsive disorder in the perinatal period: Epidemiology, phenomenology, pathogenesis, and treatment. [Trastorno obsesivo-compulsivo durante el período perinatal: epidemiología, fenomenología, etiopatogenia y tratamiento].* Anales de Psicología, 2015. **31**(1).
20. Challacombe, F.L. and A.L. Wroe, *A hidden problem: consequences of the misdiagnosis of perinatal obsessive-compulsive disorder.* Br J Gen Pract, 2013. **63**(610): p. 275-6.
21. Osnes, R.S., et al., *Mid-pregnancy insomnia is associated with concurrent and postpartum maternal anxiety and obsessive-compulsive symptoms: A prospective cohort study.* J Affect Disord, 2020. **266**: p. 319-326.
22. Cook, N., S. Ayers, and A. Horsch, *Maternal posttraumatic stress disorder during the perinatal period and child outcomes: A systematic review.* J Affect Disord, 2018. **225**: p. 18-31.
23. Facco, F.L., et al., *Sleep disturbances in pregnancy.* Obstet Gynecol, 2010. **115**(1): p. 77-83.
24. Signal, T.L., et al., *Sleep duration and quality in healthy nulliparous and multiparous women across pregnancy and post-partum.* Aust N Z J Obstet Gynaecol, 2007. **47**(1): p. 16-22.
25. Sharkey, K.M., T.B. Pearlstein, and M.A. Carskadon, *Circadian phase shifts and mood across the perinatal period in women with a history of major depressive disorder: a preliminary communication.* J Affect Disord, 2013. **150**(3): p. 1103-8.
26. Gallaher, K.G.H., et al., *The Role of Circadian Rhythms in Postpartum Sleep and Mood.* Sleep Med Clin, 2018. **13**(3): p. 359-374.
27. Foley, D., et al., *Sleep disturbances and chronic disease in older adults: results of the 2003 National Sleep Foundation Sleep in America Survey.* J Psychosom Res, 2004. **56**(5): p. 497-502.
28. Hall, M.H., et al., *Self-reported sleep duration is associated with the metabolic syndrome in midlife adults.* Sleep, 2008. **31**(5): p. 635-43.
29. Gottlieb, D.J., et al., *Association of usual sleep duration with hypertension: the Sleep Heart Health Study.* Sleep, 2006. **29**(8): p. 1009-14.
30. Cunningham, J.E.A., et al., *Acute Sleep Restriction Has Differential Effects on Components of Attention.* Front Psychiatry, 2018. **9**: p. 499.
31. Van Dongen, H.P., et al., *The cumulative cost of additional wakefulness: dose-response effects on neurobehavioral functions and sleep physiology from chronic sleep restriction and total sleep deprivation.* Sleep, 2003. **26**(2): p. 117-26.
32. Spiegel, K., R. Leproult, and E. Van Cauter, *Impact of sleep debt on metabolic and endocrine function.* The Lancet, 1999. **354**(9188): p. 1435-1439.
33. Spiegel, K., J.F. Sheridan, and E. Van Cauter, *Effect of sleep deprivation on response to immunization.* JAMA, 2002. **288**(12): p. 1471-2.
34. Reimer, M.A. and W.W. Flemons, *Quality of life in sleep disorders.* Sleep Med Rev, 2003. **7**(4): p. 335-49.
35. Fairclough, S.H. and R. Graham, *Impairment of driving performance caused by sleep deprivation or alcohol: a comparative study.* Hum Factors, 1999. **41**(1): p. 118-28.
36. Cai, S., et al., *Sleep Quality and Nocturnal Sleep Duration in Pregnancy and Risk of Gestational Diabetes Mellitus.* Sleep, 2017. **40**(2).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

37. Gelaye, B., et al., *Association of poor subjective sleep quality with suicidal ideation among pregnant Peruvian women.* Gen Hosp Psychiatry, 2015. **37**(5): p. 441-7.
38. Micheli, K., et al., *Sleep patterns in late pregnancy and risk of preterm birth and fetal growth restriction.* Epidemiology, 2011. **22**(5): p. 738-44.
39. Okun, M.L., et al., *Poor sleep quality increases symptoms of depression and anxiety in postpartum women.* J Behav Med, 2018. **41**(5): p. 703-710.
40. Lawson, A., et al., *The relationship between sleep and postpartum mental disorders: A systematic review.* J Affect Disord, 2015. **176**: p. 65-77.
41. Bei, B., et al., *Subjective perception of sleep, but not its objective quality, is associated with immediate postpartum mood disturbances in healthy women.* Sleep, 2010. **33**(4): p. 531-8.
42. Tham, E.K., et al., *Associations between poor subjective prenatal sleep quality and postnatal depression and anxiety symptoms.* J Affect Disord, 2016. **202**: p. 91-4.
43. Vasdev, S., J. Cunningham, and C. Shapiro, *Sleep Changes in PTSD*, in *Sleep and Combat-Related Post Traumatic Stress Disorder*, E. Vermetten, A. Germain, and T.C. Neylan, Editors. 2018, Springer Science+Business Media LLC: New York, NY. p. 201-208.
44. Paulson, J.L. and L. Miller-Graff, *Prenatal sleep quality and mental health symptoms across the perinatal period: A longitudinal study of high-risk women.* J Psychosom Res, 2019. **116**: p. 31-36.
45. Krawczak, E.M., et al., *Do changes in subjective sleep and biological rhythms predict worsening in postpartum depressive symptoms? A prospective study across the perinatal period.* Arch Womens Ment Health, 2016. **19**(4): p. 591-8.
46. Moher, D., et al., *Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement.* PLoS Med, 2009. **6**(7): p. e1000097.
47. Buysse, D.J., et al., *The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research.* Psychiatry Res, 1989. **28**(2): p. 193-213.
48. Bastien, C.H., A. Vallieres, and C.M. Morin, *Validation of the Insomnia Severity Index as an outcome measure for insomnia research.* Sleep Medicine, 2001. **2**: p. 297-307.
49. Giglio, L.M., et al., *Development and use of a biological rhythm interview.* J Affect Disord, 2009. **118**(1-3): p. 161-5.
50. Higgins, J.P., et al., *The Cochrane Collaboration's tool for assessing risk of bias in randomised trials.* BMJ, 2011. **343**: p. d5928.
51. Furtado, M., et al., *Risk factors of new onset anxiety and anxiety exacerbation in the perinatal period: A systematic review and meta-analysis.* J Affect Disord, 2018. **238**: p. 626-635.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

APPENDIX 1

Table 1: Key concepts, terms, and example search strategy for MEDLINE database search.

Key Concept	Perinatal Period	Anxiety	Sleep and Biological Rhythms
Free-text terms	• Perinatal • Postpartum • Post-partum • Post-partum period • Postnatal • Perinatal period • Peripartum • Pregnan* • Antenatal • Gestation • Parity • Gravid* • Puerperium	• PTSD • Post-traumatic Stress disorder • Post Traumatic Stress Disorder • GAD • Generalized Anxiety Disorder • General Anxiety • OCD • Obsessive compulsive disorder • Obsessive-compulsive disorder • Perinatal Anxiety • Anxiety • Social anxiety • Panic Disorder • Phobia	• Insomnia • Circadian • Sleep phase advance • Sleep phase delay • Sleep loss • Sleep difficult* • Sleep reactiv* • Maternal sleep • Nyctohemeral • Nychtohemeral • Diurnal • twenty-four hour rhythm • Twenty four hour rhythm • 24 hour rhythm • day night • Actiwatch • biological rhythm*
MeSH Subject Headings	• Pregnancy • Pregnancy, High-Risk • Pregnancy, Multiple • Pregnancy, Unplanned • Pregnancy, Unwanted • Gravidity • Parity • Parturition • Peripartum Period • Postpartum Period • Mothers • Single Parent • Pregnant Women • Child Rearing • Maternal-Fetal Relations	• Obsessive Behavior • Stress, Psychological • Hoarding • Impulsive Behavior • Compulsive Behavior • Anxiety • Catastrophization • [Explode] Anxiety Disorders • "Trauma and Stressor Related Disorders" • Adjustment Disorders • Stress Disorders, Traumatic • Stress Disorders, Post-Traumatic	• Chronobiology Disorders • Sleep Disorders, Circadian Rhythm • Fatigue Syndrome, Chronic • Restless Legs Syndrome • [Explode] Sleep Wake Disorders • Sleep Hygiene • [Explode] Chronobiology Phenomena • [Explode] Sleep • [Explode] Accelerometry

For peer review only

	• Puerperal Disorders	• Stress Disorders, Traumatic, Acute • Hoarding Disorder	• [Explode] Chronotherapy • Phototherapy • Sleep Phase Chronotherapy
Example Search Strategy	• Terms within key concepts combined with OR • Key concept groups of terms combined with AND • Exam*ple (truncated for clarity)*: (Perinatal OR Postpartum OR …) AND (PTSD OR Anxiety OR …) AND (Insomnia OR Circadian OR …)		

For peer review only

Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	1
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments			

For peer review only

	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	12
Sponsor	#5b	Provide name for the review funder and / or sponsor	12
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	12
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	13
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	6

For peer review only

		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	7
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	7
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	7
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	7
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	7
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	7
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	7

The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist was completed on 09. November 2020 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

For peer review only